

Figure 5: Hyperparameter sensitivity evaluation.

resentation. SComGNN *w/o m* is the variant without the mid-frequency GCN filter and the two-stage attention mechanism, thus the model only obtains the low-frequency representation. SComGNN *w/o a* is the variant without the two-stage attention mechanism thus the mid-frequency and low-frequency representations are simply concatenated. Due to similar trends and page limitations, we only show results on two metrics in Table 3, and complete results on three metrics can be found in supplementary files.

In Table 3, we can observe that (1) SComGNN achieves the best performance among the four models, indicating the collective importance of all three modules. (2) Compared to SComGNN *w/o l*, SComGNN *w/o m* achieves better performance, which verifies our observations that the low-frequency component is greater than the mid-frequency component. (3) In some cases, the performance of SComGNN *w/o a* can even be inferior to that of SComGNN *w/o m*. This indicates although the mid-frequency representation is valuable, its integration needs to be more efficient.

### Hyperparameter Analysis

We investigate the impact of two hyperparameters on the performance of our model, i.e., the depth of low-pass and mid-pass GCN layers and the embedding size. Due to similar trends and page limitations, we only present the NDCG results in Figure 5, and complete results can be found in supplementary files. We can make some conclusions. First, for the depth of GCN layers, the performance may decrease as the depth of network layers increases. This is because our model aggregates structural information from different perspectives so that a one-layer network can perform well. Next, for the embedding size, 16 is the most appropriate value. Increasing the embedding size to 32 and 64 not only does not necessarily improve the model performance but also increases the model complexity and training time. Also, reducing the embedding size to 8 or 4 results in a significant drop in performance, indicating the importance of learning rich and expressive feature representations.

### Case Study

To assess the impact of low and mid-frequency components in the production environment, we compare the performance of three models: SComGNN *w/o m*, SComGNN *w/o l*, and SComGNN. Figure 6 shows the TOP-3 complementary items recommended for “Instant Coffee”. In the recommendations solely based on the low-frequency components, items are all coffee, which is strongly similar to

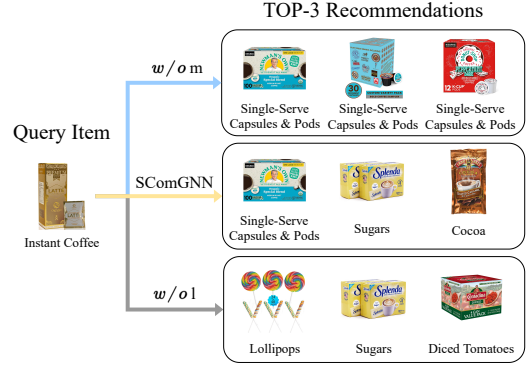


Figure 6: Examples of complementary recommendation results with different frequency components.

the query item. Conversely, with only the mid-frequency components, items appear to have a low correlation to the query item. However, with both the low-frequency and mid-frequency components, a diverse range of items, including coffee, sugar, and cocoa, are recommended. The inclusion of sugar as a flavoring for coffee, and the presence of cocoa alongside coffee as distinct but related beverages, illustrate our ability to capture both relevance and dissimilarity. This outcome highlights the crucial roles played by the low-frequency component in representing relevance and the mid-frequency component in representing dissimilarity, both of which are essential for complementary relationships.

### Conclusion

In this paper, we bridge the gap between spectral properties and attributes of complementary relationships. Our analysis reveals that complementary item graphs primarily consist of low-frequency and mid-frequency components in the spectral domain, representing relevance and dissimilarity attributes. We propose GCN filters to extract the two components and employ a two-stage attention mechanism for adaptive integration. Experiments on four publicly available datasets demonstrate the effectiveness of our theoretical analysis and the proposed method. In the future, more effective GCN filters and integration approaches of two frequency components deserve to be explored.

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